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Adaptive undersampling and short clip-based two-stream CNN-LSTM model for surgical phase recognition on cholecystectomy videos

Sang-Goo Lee [a,c], Ga-Young Kim [b], Yoo-Na Hwang [a,c], Ji-Yean Kwon [a,c], Sung-Min Kim [a,c,*]

[a] Department of Medical Device and Healthcare, Dongguk University-Seoul, (04620) 30, Pildong-ro-1-gil, Jung-gu, Seoul, Republic of Korea
[b] Radiation Oncology and Molecular Radiation Sciences, Johns Hopkins University School of Medicine, USA
[c] Department of Regulatory Science for Medical Device, Dongguk University-Seoul, (04620) 30, Pildong-ro-1-gil, Jung-gu, Seoul, Republic of Korea

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ABSTRACT

Surgical phase recognition is challenging due to overfitting problems caused by imbalanced data among surgical phases. We proposed an adaptive sampling rate-based undersampling method that could generate the number of each surgical phase data similarly to alleviate biased learning. To improve the performance of our method, we also introduced a two-stream CNN-LSTM model that could extract temporal information on behavioral changes between each image frame. First, we extracted a total of 40,236 short clips using an adaptive subsampling rate from the entire video. Each short clip was entered into a pre-trained GoogLeNet. The output with visual information was then immediately fed into a sequence-to-sequence LSTM model to extract temporal information of neighbor frames within a short clip. At the same time, another sequence-to-vector LSTM was used, to extract temporal information from all successive image frames to predict the final surgical phase. The proposed method was evaluated with a public dataset Cholec80. The proposed approach outperformed state-of-the-art methods, showing a high F1-score of 87.12% and an AUC of 98.00%. In addition, the F1-score deviation between all phases decreased by about 10% compared to that before applying undersampling. Experimental results confirmed that employing our proposed method could learn enrich temporal information from short clips. It outperformed the conventional one-stream CNN-LSTM architecture.

1. Introduction

Minimally invasive surgery(MIS) [1,2] is a method that can minimize wounds or scars on the human body by reducing the incision area during laparotomy. MIS is becoming common due to its many advantages such as less amounts of bleeding and scarring, a short recovery period, and a less risk of infection. A typical MIS is a cholecystectomy. About 90% of cholecystectomies are performed laparoscopically in the United States [3]. With increasing interest in MIS, interest in recognizing surgical workflow from MIS video data is also increasing.

Recognition of surgical workflow, particularly, for phase recognition [4,5], tool detection [6–8], and event recognition [9], is a key factor to support surgical teams and optimize patient treatment inside an operating room. In cholecystectomy, surgical phase recognition is very important for optimizing surgical workflow, training less experienced surgeons, and providing intraoperative assistance [10,11]. Surgical phase recognition can identify different phases of an entire cholecystectomy surgical procedure, such as preparation, Calot's triangle dissection, clipping cutting, gallbladder dissection, gallbladder packaging, cleaning coagulation, and gallbladder retraction.

2. Related work

Hidden Markov Model (HMM) and artificial neural network are the most commonly used machine learning(ML) models [12]. Previous studies for surgical phase recognition have investigated the use of a sensor-based signal source as input data [13–16]. The majority of previous studies have used video data or tool binary signals. Dynamic time warping (DTW) and HMM were the two most commonly used models for phase recognition from tool binary signals [17]. These models mainly could help us learn the temporal information of surgery. Hierarchical HMM has been combined with other ML models such as convolutional neural networks (CNNs) to improve accuracy [14,18]. However, these signals are usually obtained via manual annotation of surgical videos or

* Corresponding author at: Department of Medical Device and Healthcare, Dongguk University-Seoul, (04620) 30, Pildong-ro-1-gil, Jung-gu, Seoul, Republic of Korea.

E-mail address: smkim@dongguk.edu (S.-M. Kim).

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by installing additional sensors. Many studies have focused on using visual features by integrating convolutional neural networks (CNNs) with recurrent neural networks (RNNs) or long short-term memory (LSTM) [5,19–21]. Although previous studies have used visual features with surgical tool information such as type or presence, annotation is time-consuming which is a disadvantage. To overcome annoying tasks using only surgical phase information, various strategies have been proposed, such as prior knowledge inference (PKI) and remaining surgery duration (RSD) [22–24]. To reduce training time, a temporal convolutional network (TCN) [25] and a transformer network [26] instead of RNN have been used. Transformer networks can leverage temporal relationships between current and previous frames using self-attention, similar to the LSTM method. However, existing works on surgical phase recognition from intra-operative videos remain challenging even for advanced CAI systems of cholecystectomy due to problems such as variability of patient anatomy, different surgeon style, and class imbalances [27], which can lead to degradation of performance It is important to handle the problem of class imbalance in multi-class classification because classes with relatively fewer data are more difficult to learn than other classes. However, to the best of our knowledge, none of the studies using the Cholec80 [14] dataset dealt with the class imbalance problem. The aim of this study was to recognize the phase in laparoscopic videos for cholecystectomy procedures for gallbladder resection. We introduced adaptive temporal subsampling-based two-stream CNN-LSTMs by learning the temporal relationships between long sequences and short clips from entire video sequences.

Our main contributions were two-fold as follows: (1) we tackled the class imbalance problem in surgical phase recognition. We proposed a novel undersampling approach that could alleviate biased learning using different sampling rates (frames per second or fps) for each phase; and (2) we introduced two-stream CNN-LSTMs, with CNN for extracting visual features and two-stream LSTMs for extracting temporal contextual information. One of these two-stream LSTMs was the LSTM (seq2-seq) model that was trained with the aim of predicting the phase in a frame using the sequence of previous neighboring frames for short-range contextual information. The other one was the LSTM (seq2vec) model which was trained to recognize the phase in the last frame of every clip for a long-term view. The remainder of this paper was organized as follows. Section 3 introduces our methods. Sections 4 and 5 provide experimental results and discussions, respectively. Section 6 concludes the paper and suggests future studies.

3. Materials and methods

An overview of our proposed methods and training process is presented in Fig. 1. We proposed an adaptive undersampling strategy that could decrease biased learning by creating the number of short clips between all of the seven classes similarly. We also exploited a GoogLeNet to extract visual features and utilized two LSTMs to optimize the temporal context information from short clips of sequential images.

3.1. Dataset

This study used the publicly available Cholec80 dataset containing 80 cholecystectomy videos. Thirteen surgeons performed these 80 cholecystectomies. All surgical videos were recorded at 25 fps with a resolution of 1920 × 1080 or 854 × 480 pixels. In the Cholec80 dataset, videos were captured at 25 fps with annotations for surgical phases, whereas surgical instruments were annotated at 1 fps by experienced surgeons. Table 1 shows an overview of all data with the total number of frames and total duration for each phase.

Table 1
Compositions of Cholec80 dataset.

Phase	Action	Total Frames (Total duration(s))
1	Preparation	214,300 (8,572)
2	Calot's Triangle Dissection	1,842,525 (73,701)
3	Clipping Cutting	352,000 (14,080)
4	Gallbladder Dissection	1,460,825 (58,433)
5	Gallbladder Packaging	190,450 (7,618)
6	Cleaning Coagulation	485,650 (19,426)
7	Gallbladder Retraction	165,925 (6,637)
	TOTAL	**4,711,675 (188,467)**

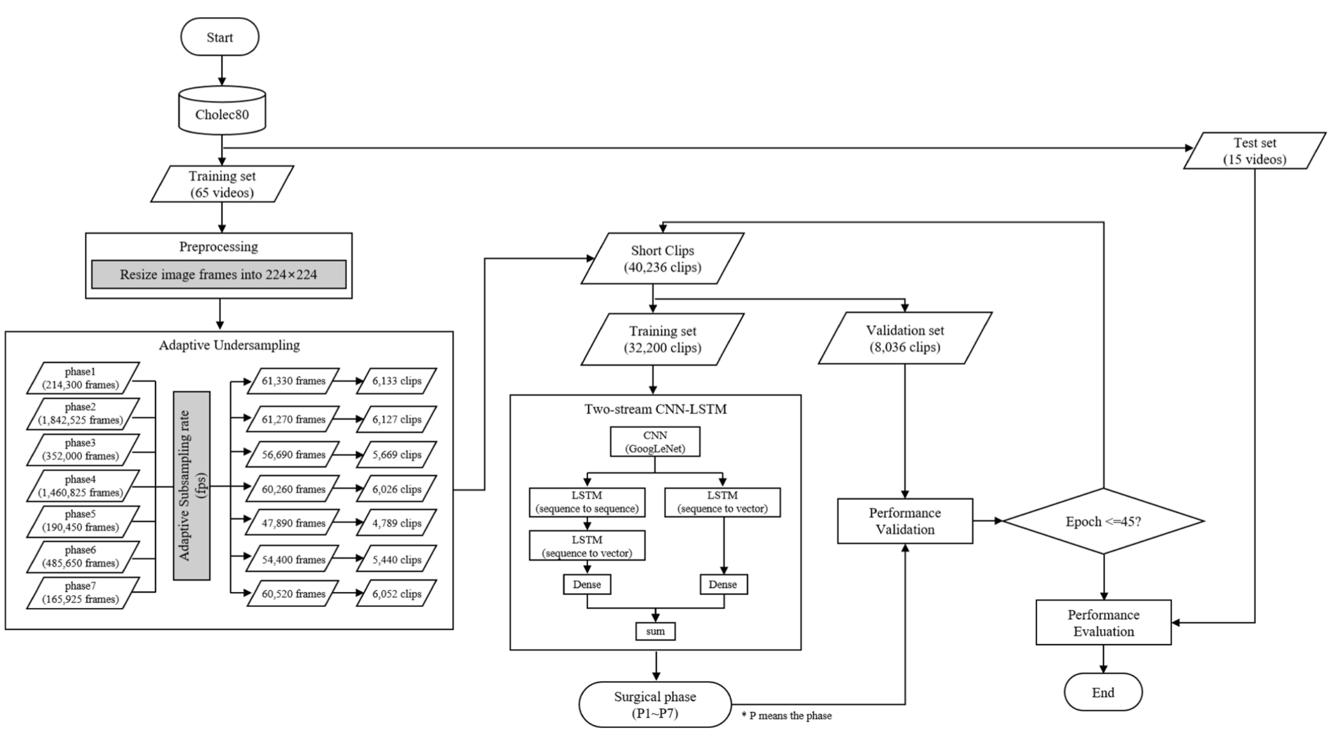


Fig. 1. A flowchart showing the proposed framework of surgical phase recognition.

3.2. Adaptive sampling rate-based undersampling method

Each surgical phase generally had different fine-grained actions. Extracting different motion information for each phase from the surgical video is crucial for accurate recognition. Table 2 (a) shows total number of datasets subsampled to 1 fps in Cholec80. Phases 2 and 4 accounted for about 68% of all clips of the training set. This data imbalance of each phase could lead to biased learning and affect performance negatively. Therefore, we proposed a novel undersampling method using short clips with adaptive temporal subsampling.

When dealing with challenges, we subsampled the entire surgical video in short clips which consisted of 10 frames with different sampling rates for each phase. Each short clip was assigned to one of seven classes corresponding to each surgical phase. To capture both spatial and temporal actions of an intra-operative video clip, we focused on a model that could learn features of a group of frames rather than a single image frame. Surgical videos for training were subsampled as shown in Table 2 (b). To generate the number of clips similarly for the seven surgical phases, we used adaptive-frame clips subsampled by 8.33, 1, 5, 1.25, 8.33, 4.17, and 12.5 fps for each phase. The fps for each phase were determined by calculating the stride for each phase. The image frames of the clips for each phase were subsampled based on the stride (Fig. 2). The stride was calculated using the following equation (1):

$$stride = \frac{P}{M} \times 25 \tag{1}$$

where stride implies a sampling interval that is rounded to the nearest whole number, P represents the number of image frames in each phase, and M indicates the highest number of image frames across all phases. Consequently, the stride for the remaining classes was determined based on the class with the largest amount of data. In the case of the Cholec80 dataset, the second phase had the highest number of image frames. Therefore, we set the stride for the second phase to 25(equivalent to 1 fps) and adjusted the stride for the remaining phases to values lower than 25 to ensure a similar number of clips in each phase. For example, in phase 7, ten continuous frames with a stride of 2 were extracted to generate each short clip, meaning that all even-numbered frames were extracted from 165,925 frames of phase 7, resulting in a total of 6,052 short clips. During subsampling, we excluded the remaining frames when a short clip could not be created. As a result, we obtained a total of 40,236 short clips (6,133, 6,127, 5,669, 6,026, 4,789, 5,440, and 6,052 clips for the 7 phases, respectively) from the original Cholec80 dataset.

3.3. Two-stream CNN-LSTM architecture

To extract both visual and temporal information from surgical action, most of the previous models were designed with CNN-RNN. Each RNN was trained on entire videos to extract long-temporal context information such as surgical tools and surgical phases. However, this approach has an overfitting problem because of class imbalanced videos having different lengths for each phase. We proposed two-stream LSTM structures following the pretrained GoogLeNet (Fig. 3). The last classifier layer of GoogLeNet was replaced with our two-stream LSTM model. Outputs from the last inception module of GoogLeNet were vectorized by global max pooling 2D layer. Finally, Visual feature vectors obtained from GoogLeNet were inputted into the two-stream LSTMs for capturing the time relationship. One LSTM stream was a sequence-to-vector model that took long-term information from short clips. Another stream was a sequence-to-sequence LSTM model that captured short-term information from frames in each clip. These two LSTMs could cooperatively extract surgical action information for not only the entire motion within a single short clip but also between each frame. Finally, the model determined the current surgical phase after prediction scores of two-stream LSTMs were combined using an add function.

In the training procedure, GPU could not hold a large video frame data load of about 160 GB at once. Thus, we repeated the data loading process per epoch until all training and validation videos had been trained. Our framework was conducted in Python 3.6.12, using a Tensorflow-GPU 2.4.0 environment, with an NVIDIA GeForce RTX 3090. Hyper-parameters used in the network were as follows.

- Optimizer: Adam
- Momentum: 0.9
- Dropout rate of LSTM: 0.25
- Initial learning rate: 0.00001(divided by 2 at 15 epochs)
- Epochs: 45
- Activation function of LSTM: sigmoid
- Loss function: categorical cross-entropy
- Size of the batch: 16

3.4. Evaluation matrix

The Cholec80 dataset was divided at a ratio of 8:2 for training and testing sets. The proportion of the validation set was 20 % of the training set. To verify the performance of phase recognition in the Cholec80 dataset. Descriptive statistics (mean ± standard deviation) was performed for data obtained from the testing set. We aggregated the confusion matrix of each test video and measured accuracy (ACC), precision (PR), negative predictive value (NPV), recall (RE), specificity (SP), F1-score, and area under the curve (AUC) [28]. In particular, the F1-score is affected by data distribution for each category [28]. It was calculated as follows:

$$F1 - score = \frac{2TP}{2TP + FP + FN} \tag{2}$$

where TP, FP, and FN were the total number of true positive, false positive, and false negative classes in a short clip, respectively. Since we performed seven-class classification tasks, total TP, FP, and FN were computed as the sum of TP, FP, and FN for seven phases. True positive for each phase was the number of clips where both the label and the model prediction were positive. False positive for each phase was the number of clips where only the model prediction was positive. False negative was the number of clips where only the label was positive.

4. Experimental result

4.1. Comparison results of phase recognition performance among different CNN models

The proposed model consisted of three main components: a CNN model for visual feature extraction and two-stream LSTMs for integrating temporal information. To decide the optimal CNN architecture

Table 2
Number of clips with different subsampling rates for phase recognition.

Group		Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7
(a) Previous subsampling method	Subsampling rates (fps)	1	1	1	1	1	1	1
	Sequences per phase (clips)	1,381	5,755	1,457	4,664	735	925	437
(b) Proposed method (adaptive temporal-subsampling)	Subsampling rates (fps)	8.33	1.00	5.00	1.25	8.33	4.17	12.50
	Sequences per phase (clips)	6,133	6,127	5,669	6,026	4,789	5,440	6,052

fps: Frame per second.

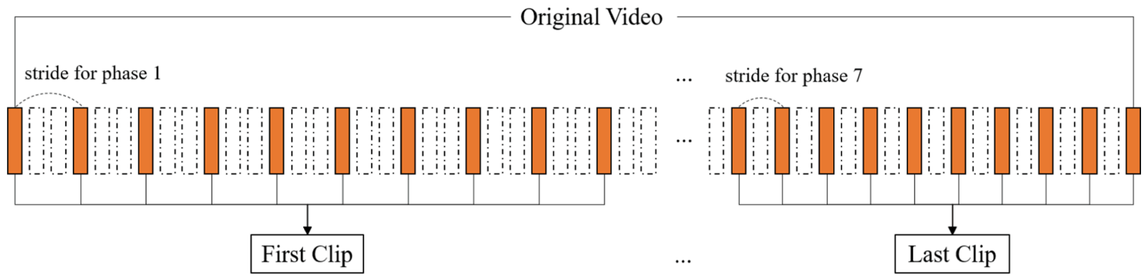


Fig. 2. The method of undersampling to obtain image frames of clips for each phase using the stride from the surgical video.

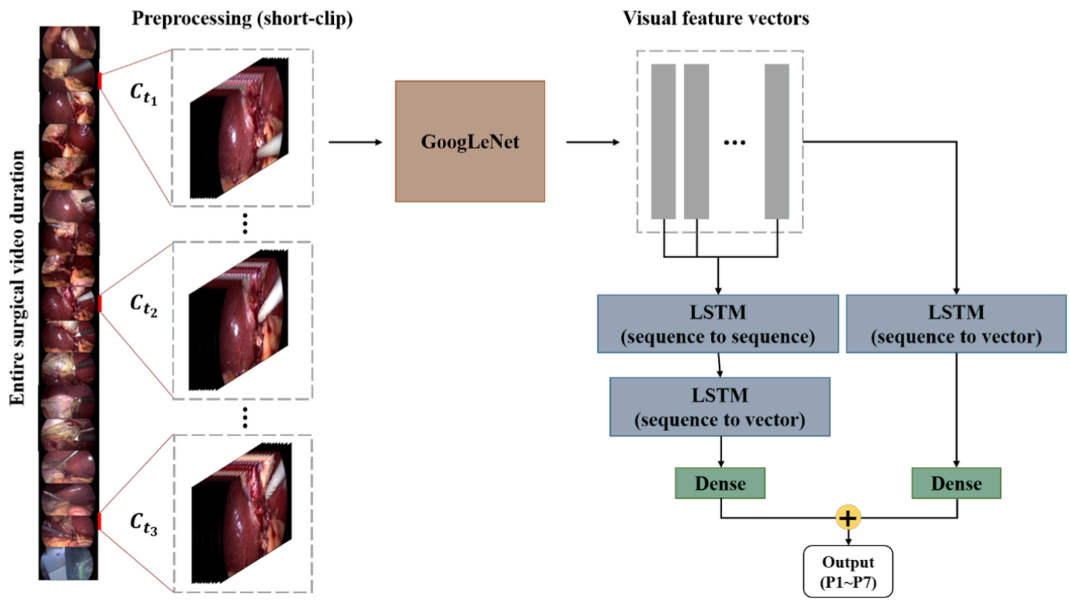


Fig. 3. Architecture of the proposed CNN-LSTM phase recognition model. Input clip is entered into a pre-trained GoogLeNet. The output of visual features is vectorized by the global max pooling layer. Two LSTMs capture the temporal context using the sequential visual feature vectors within the clip to identify the surgical movements associated with a particular phase. C_{t_1}, C_{t_2}, C_{t_3} means the clip at a specific time in the entire surgical video duration.

of our final model, we compared the phase recognition performance of VGG16, ResNet50, and GoogLeNet pretrained with ImageNet data [29] in the empirical experiment (Table 3). We found that the optimal CNN model of our two-stream CNN-LSTMs was GoogLeNet because it showed a higher F1-score (76.79%) than VGG16 (75.24%) and ResNet50 (76.02%). In terms of all other metrics, GoogLeNet also outperformed VGG16 and ResNet50. Furthermore, the standard deviation of accuracy was 8.54% and 10.42% lower than that of VGG16 and ResNet50, respectively. These results indicate that GoogLeNet could extract more suitable features from the single frames of each phase.

4.2. Effect of the number of frames contained in a single clip

For phase recognition based on short clips, it is necessary to select the number of frames included in a single clip. We compared the run-time and F1-score when the number of frames in a single clip was set to 5, 7, 10, 15, or 20. All comparisons were computed with the same dataset under the same experimental environment. The comparison results are

Table 4
Comparison results of accuracy, F1-score, AUC, and run-time among different numbers of frames on a single clip.

Short clip-based phase recognition				
The number of frames in a single clip	ACC (%)	F1-score (%)	AUC (%)	Runtime per epoch (Minutes)
5	83.18 ± 8.66	75.22 ± 13.43	95.60 ± 4.60	26
7	81.55 ± 29.29	73.33 ± 14.66	95.60 ± 6.60	23
10	**84.46 ± 10.54**	**76.79 ± 11.58**	**96.33 ± 4.33**	**21**
15	81.02 ± 18.59	73.84 ± 12.07	96.07 ± 4.07	21
20	85.28 ± 19.98	75.57 ± 9.80	94.80 ± 9.80	19

ACC: Accuracy and AUC: Area under the ROC curve.

Table 3
Comparison results of overall phase recognition performance according to different CNN architectures.

CNN	ACC (%)	PR (%)	NPV (%)	RE (%)	SP (%)	F1-score (%)	AUC (%)
VGG16	83.62 ± 19.08	75.76 ± 13.90	96.85 ± 3.81	75.28 ± 16.97	96.80 ± 4.69	75.24 ± 14.70	96.00 ± 5.00
ResNet50	82.45 ± 20.96	75.65 ± 15.51	96.49 ± 8.16	76.42 ± 11.54	96.61 ± 6.54	76.02 ± 11.63	95.80 ± 7.80
GoogLeNet	**84.46 ± 10.54**	**77.95 ± 13.70**	**96.98 ± 5.67**	**76.50 ± 15.83**	**96.86 ± 8.30**	**76.79 ± 11.58**	**96.33 ± 4.33**

CNN: Convolutional neural network, ACC: Accuracy, PR: precision, NPV: Negative predictive value, RE: Recall, SP: Specificity, and AUC: Area under the ROC curve.

summarized in Table 4.

The experimental results showed that the highest F1-score was 76.79% when the number of frames in a single clip was 10. It was 1.57%, 3.46%, 2.95%, or 1.22% higher than when the number of frames was set to 5, 7, 15, or 20 respectively. The AUC was also the highest (96.33%) when the number of frames in a single clip was 10. When the number of frames in a clip was set to 10, the accuracy decreased by 0.82% compared to when it was set to 20. However, it is worth noting that the standard deviation was 9.44% lower in the case of 10 frames. The runtime per one epoch had the following relationship: 5 frames (26 min) > 7 frames (23 min) > 10 or 15 frames (21 min) > 20 frames (19 min). As the number of image frames within a single clip increased, less training time was needed. However, it decreased the total amount of trainable data. Thus, the optimal number of frames in each short clip was set to 10, considering its outstanding classification ability, statistical results, and training time.

4.3. Effect of adaptive temporal subsampling on data imbalance problem

To prove the effectiveness of our adaptive temporal subsampling approach for all phases, we compared the F1-score and AUC of phase recognition results before and after applying the proposed method. Group (1) was downsampled to 1 fps for all phases, resulting in a total of 15,354 clips. This downsampling approach of Group (1) is generally used for phase recognition of surgical video because of its low computational time for training. Other groups (Group (2), Group3, and Group 4) were composed of a total of 67,192, 40,236, and 25,119 clips, respectively. Our adaptive temporal subsampling approach was applied to these groups. Comparative experiments were performed under the same conditions for all groups. GoogLeNet was used as a CNN. A single clip was composed of 10 frames. Transfer learning was then applied. Results of different groups are shown in Table 5.

Overall F1-score and AUC of Group 3 were remarkably improved (increases of 10.33% and 1.67%, respectively) compared to those of Group (1). Group 3 showed higher AUCs (over 95%) in all phases than Group (1). In phases 1, 3, 5, 6, and 7, the F1-scores of Group 3 were greatly increased by 12.01%, 9.65%, 13.43%, 10.84%, and 14.60%, respectively, than those of Group (1) which showed an imbalanced data problem. By applying our adaptive temporal subsampling approach, F1-scores for all phases were improved in Group 3. However, phase 1 and phase 7 showed lower F1-scores (8.56% and 3.47% lower, respectively) than the overall F1-score of Group 3 (87.12%).

A two-sample *t*-test was performed to test differences in standard deviations of F1-score and AUC between before and after applying the adaptive temporal subsampling approach in phases 1, 3, 5, 6, and 7 which had relatively less numbers of frames among the seven cholecystectomy phases. To that end, we compared Group (1) and Group 3 because Group (1) had an equal subsampling rate of 1 fps in all phases, and Group 3 had the highest performance of our approach as shown in the experiment described above (section 3.3). Fig. 4 and Fig. 5 show *t*-test results of the F1-score and AUC for phase 1, phase 3, phase 5, phase 6, and phase 7. Smaller p-values meant that differences between the two groups were more remarkable.

In an ablation experiment before and after applying our undersampling approach, F1-scores of phase 3, phase 5, and phase 7 demonstrated significant differences between Group1 and Group 3 ($p < 0.05$). Especially, our approach for phase 5 resulted in a relatively higher increase in the F1-score than before, showing a statistically significant difference ($p < 0.01$). After solving the imbalanced data structure, results of AUC of phases 5 and 6 also showed statistically significant differences between Group1 and Group 3 ($p < 0.05$).

Results of confusion matrixes before and after applying our adaptive temporal subsampling approach are illustrated in Fig. 6. A 7 × 7

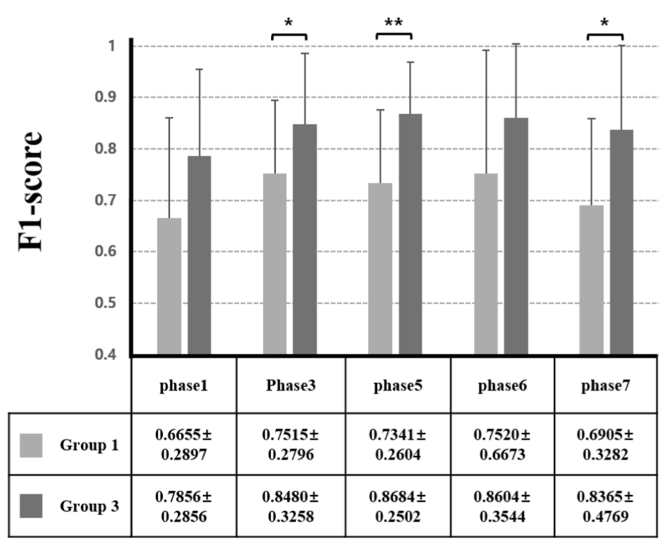
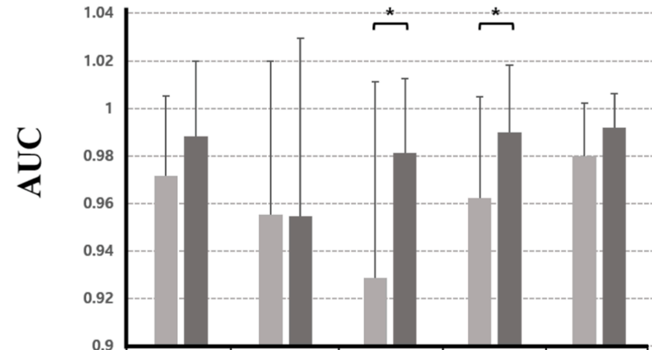


Fig. 4. F1-score trends before and after applying the proposed undersampling method.

Table 5
F1-score and AUC of phase recognition results for all phases according to different subsampling rates.

Group		Surgical phase on Cholec80 dataset							Total
		Phase1	Phase2	Phase3	Phase4	Phase5	Phase6	Phase7	
The number of clips	Group (1)	1,381	5,755	1,457	4,664	735	925	437	15,354
	Group (2)	10,093	9,239	9,837	9,807	7,524	9,670	11,022	67,192
	Group 3	**6,133**	**6,127**	**5,669**	**6,026**	**4,789**	**5,440**	**6,052**	**40,236**
	Group 4	3,736	3,654	3,239	3,717	3,578	3,555	3,640	25,119
F1-score (%)	Group (1)	66.55 ± 28.97	87.62 ± 33.14	75.15 ± 27.96	86.43 ± 19.92	73.41 ± 26.04	75.20 ± 66.73	69.05 ± 32.82	76.79 ± 11.58
	Group (2)	64.36 ± 19.81	84.54 ± 32.34	72.55 ± 42.42	77.46 ± 58.92	70.67 ± 38.44	66.23 ± 43.30	56.95 ± 41.39	70.07 ± 17.40
	Group 3	**78.56 ± 28.56**	**92.08 ± 14.71**	**84.80 ± 32.58**	**92.20 ± 13.69**	**86.84 ± 25.02**	**86.04 ± 35.44**	**83.65 ± 47.69**	**87.12 ± 6.30**
	Group 4	68.06 ± 30.96	82.96 ± 17.13	72.43 ± 52.22	75.91 ± 58.99	69.68 ± 30.60	73.24 ± 46.57	59.21 ± 31.70	73.84 ± 12.19
AUC (%)	Group (1)	97.15 ± 2.85	97.53 ± 3.53	95.53 ± 15.53	96.47 ± 7.47	92.87 ± 27.87	96.21 ± 8.21	98.00 ± 6.00	96.33 ± 4.33
	Group (2)	96.69 ± 6.69	92.80 ± 18.80	95.33 ± 13.33	94.93 ± 12.93	96.73 ± 8.73	91.25 ± 36.25	98.27 ± 3.27	95.27 ± 5.27
	Group 3	**98.83 ± 9.83**	**97.07 ± 6.07**	**95.47 ± 22.47**	**98.27 ± 6.27**	**98.13 ± 7.13**	**99.00 ± 10.00**	**99.20 ± 3.20**	**98.00 ± 5.00**
	Group 4	98.64 ± 4.64	93.73 ± 9.73	95.33 ± 27.33	92.07 ± 30.07	94.27 ± 5.73	93.86 ± 30.86	97.27 ± 62.67	95.07 ± 11.07

AUC: Area under the ROC curve.



	phase1	Phase3	phase5	phase6	phase7
Group 1	0.9715± 0.0285	0.9553± 0.1553	0.9287± 0.2787	0.9621± 0.0821	0.9800± 0.0600
Group 3	0.9883± 0.0983	0.9547± 0.2247	0.9813± 0.0713	0.9900± 0.1000	0.9920± 0.0320

Fig. 5. AUC trends before and after applying the proposed undersampling method.

confusion matrix was generated because we performed classification for seven surgical phases. The brighter the area, the more incorrect the answer. Fig. 6 shows that the proportions of incorrect answers of our approach are relatively decreased for phases 1, 3, 5, 6, and 7 than those before our adaptive temporal subsampling approach. Consequently, we expected a positive effect from preventing biased learning by balancing the amount of data between each phase considering improvements of F1-score and AUC, small p values, and confusion matrix.

Fig. 7 and Fig. 8 show stacked bar graphs comparing model results' labels and predictions for the top two of 15 videos used as test sets. Fig. 7 shows the 44th video of the Cholec80 dataset with all seven phases present. F1-score and accuracy were 94.86% and 98.18%, respectively. Fig. 8 shows the 23rd video consisting of six phases without P1, showing an F1-score of 96.33% and an accuracy of 97.92%.

4.4. Comparisons with prior works

This study compared phase recognition performances between our model and previous models that used cholecystectomy video data as shown in Table 6. PhaseLSTM [30], EndoLSTM [30], ResNetLSTM [22], EndoN2N [24], SV-RCNet [22], and SE-OHFM [31] models with one CNN and one RNN connected in series, OperA [26] using transformer network, and TeCNO [25] with TCN were compared.

The model proposed in this study has a structure in which two LSTMs

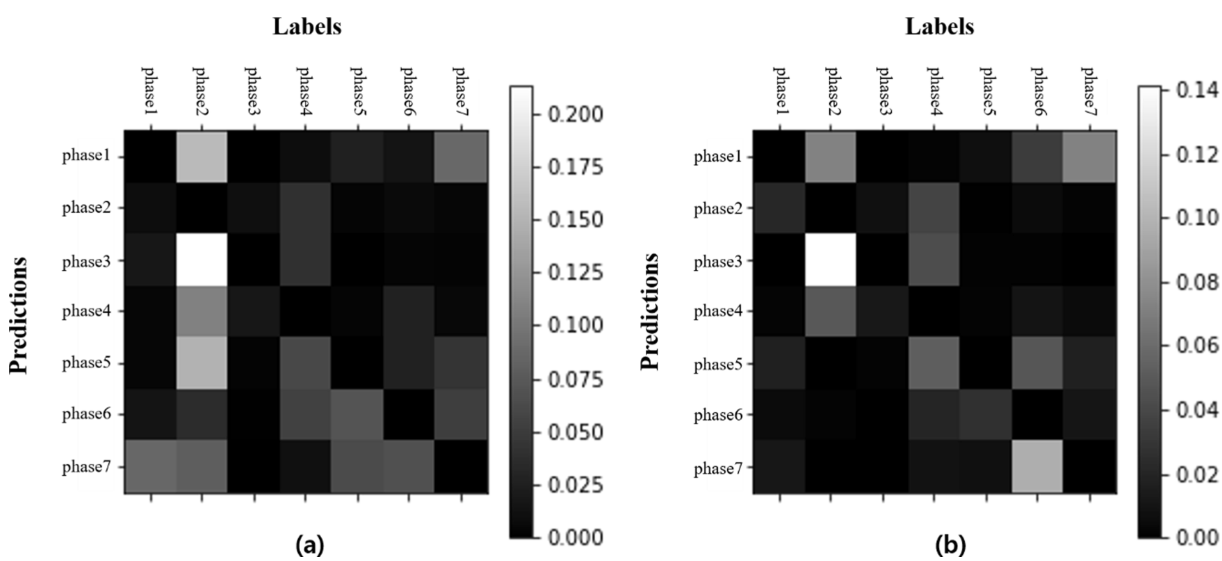


Fig. 6. (a)-Color visualizations of the confusion matrix before applying the proposed undersampling method, (b)-Color visualizations of the confusion matrix after applying the proposed undersampling method (The brighter area is the higher error rate).

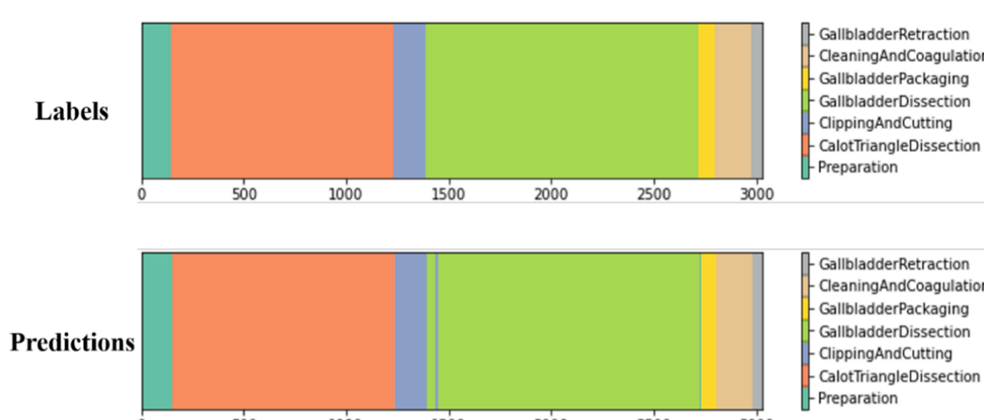


Fig. 7. Stacked bar graph of results for the 44th video in Cholec80 dataset.

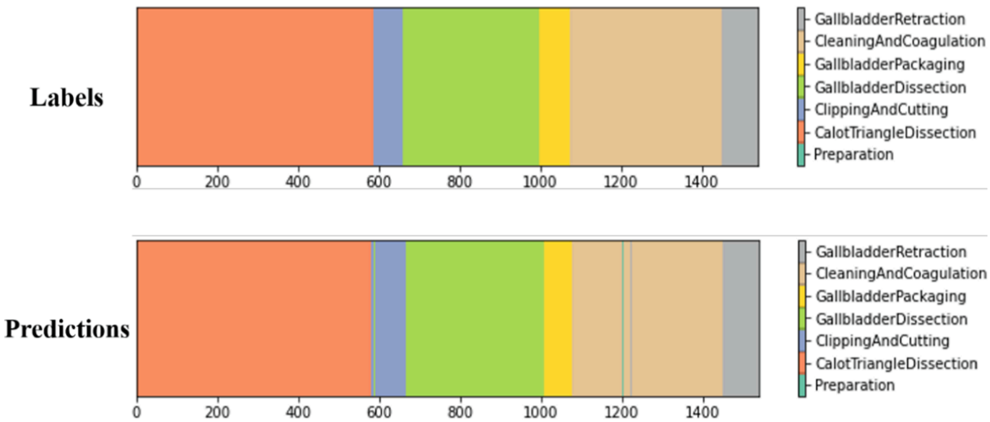


Fig. 8. Stacked bar graph of results for the 23rd video in Cholec80 dataset.

Table 6
Comparative results of phase recognition between existing and proposed methods for surgical videos from Cholec80 dataset.

Model	ACC (%)	PR (%)	RE (%)	F1-score (%)
PhaseLSTM [30]	79.68 ± 0.07	72.85 ± 0.10	73.45 ± 0.12	73.15 ± 0.11
EndoLSTM [30]	80.85 ± 0.17	76.81 ± 2.62	72.07 ± 0.64	74.36 ± 1.58
ResNetLSTM [22]	86.58 ± 1.01	80.53 ± 1.59	79.94 ± 1.79	80.23 ± 1.69
EndoN2N [24]	86.7 ± 9.30	81.4 ± 23.00	80.9 ± 22.10	81.1 ± 7.50
TeCNO [25]	88.56 ± 0.27	81.64 ± 0.41	85.24 ± 1.06	83.40 ± 0.72
OperA [26]	91.26 ± 0.64	–	–	84.49 ± 0.64
SV-RCNet [22]	92.4 ± 5.20	–	–	–
SE-OHFM [31]	85.80 ± 8.10	–	–	–
Proposed	90.47 ± 11.37	87.53 ± 11.55	87.73 ± 6.64	87.12 ± 6.30

ACC: Accuracy, PR: precision, and RE: Recall.

are connected in parallel to minimize lost temporal information. Each LSTM can learn temporal information about changes in motion between each image frame and the overall action within a short clip. F1-score was used as the main performance indicator for comparing our proposed method's performance with previous studies because F1-score is affected by the distribution of data imbalances. For OperA and EndoN2N models, their F1-scores were described in the literature. However, F1-scores for the remaining models were not mentioned in the literature. Thus, we calculated F1-scores for these models directly based on precision and sensitivity. The F1-score of the proposed final model was 87.12%. The best result was about 3% higher than OperA, the top model. Our AUC and accuracy were 98.00% and 90.47%, respectively. Since we exploited the GoogLeNet, a deep convolutional neural network with improved computational efficiency, we could experiment with relatively few parameters and epochs compared to other models despite utilizing both a CNN and two LSTMs as shown in Table 7.

5. Discussion

Surgical workflow recognition of CAI can help medical staff optimize patient treatment in the operating room. It can also be used as educational material for junior doctors. Previous studies improved surgical phase recognition performance using visual and temporal information within the video [5,19–26,30–31]. One of the main concerns of employing deep CNNs in phase recognition tasks is the problem of data imbalance between the phases of surgical videos [5,20,25,26,30–31]. To improve the performance of phase recognition in cholecystectomy videos, we proposed a two-stream CNN-LSTM based on a novel adaptive temporal subsampling by leveraging different sampling rates. We focused on recognizing the main phase from short video clips known to

Table 7
Results of comparing different traning hyperparameters for individual models.

Model	Manner	Structure	Learning rate	Epoch	Batch size (images)	Total parameter
PhaseLSTM [30]	Separated	Alexnet +LSTM	1e-3 for AlexNet (divided by 10 every 20 K epoch) 1e-2 for LSTM	50 K for CNN 30 K for LSTM	50 for CNN 3,993 for LSTM	29,982,471
EndoLSTM [30]	Separated	Alexnet +LSTM	1e-3 for AlexNet (divided by 10 every 20 K epoch) 1e-2 for LSTM	50 K for CNN 30 K for LSTM	50 for CNN 3,993 for LSTM	30,039,822
ResNetLSTM [22]	Separated	ResNet50 +LSTM	5e-4 for ResNet 5e-3 for LSTM (divided by 10 every 20 k epoch)	–	–	24,133,255
EndoN2N [24]	End-to-End	AlexNet +LSTM	1e-4 (divided by 4 every 2 k epoch)	8 K	6,000	27,971,854
TeCNO [25]	–	TCN	5e-4	25	Length of each video	–
OperA [26]	–	Transformer	1e-5	30	Length of each video	–
SV-RCNet [22]	End-to-End	ResNet50 +LSTM	5e-4 for ResNet 5e-3 for LSTM (divided by 10 every 20 k epoch)		250	23,645,336
Proposed	End-to-End	GoogLeNet +two LSTMs	1e-5 (divided by 2 at 15 epochs)	45	160	22,928,174

Total parameters for each model are computed under the same input conditions with a size of 224 × 224 pixels. (Total parameters of SV-RCNet are calculated by minimum because they are not presented a number of units of LSTM.)

be effective in very constrained environments.

Surgical videos provide valuable visual information, including the current situation (such as incision, suture, resection, etc.) and details about surgical tools (length, tip shape, quantity, etc.). In other words, both local information (fine-grained characteristics) and global information (overall context) are crucial. In the empirical experiments, GoogLeNet outperformed ResNet50 and VGG16 across all evaluation metrics. Thus, we selected GoogLeNet as the CNN for the proposed model. Notably, the accuracy of GoogLeNet had a significantly lower standard deviation than the other models. GoogLeNet uses multiple small convolutions to construct deep-structured networks with fewer parameters and computational complexity. It progressively extracts more complex features as the network depth increases from the initial layer that detects simple patterns. GoogLeNet can also use various kernel sizes (1, 3, 5, and 7) to learn both local and global features better than other models. These characteristics of GoogLeNet led to low variability and generalized surgical phase recognition performance compared to the other models.

In each surgical phase with imbalanced data, preventing overfitting for a particular phase has a positive effect on improving the model's performance [32]. In previous studies [19,24–26,30–31], models were trained by downsampling unbalanced surgical video data at the same sampling rate for each surgical phase. In contrast, we proposed an adaptive temporal subsampling approach using different sampling rates at each phase to make numbers of clips similar for all phases. We compared the performance of surgical phase recognition before and after applying our subsampling approach. Experimental results indicated meaningful differences in average F1-score and AUC after applying the proposed method. F1-scores of phase 1 (preparation) and phase 7 (retraction) were commonly lower than those of other phases in all groups, regardless whether our approach was applied. This might be due to the lack of surgical motion information and short surgical duration. When the architecture is complex and the number of training datasets is small, the underfitting problem may appear. Conversely, if there are too much data, an overfitting problem might occur. To solve this problem, the relationship between bias and variance must be considered. Due to the high complexity of the proposed architecture in this study, we increased the complexity of data by setting sampling rates higher for phases 1, 3, 5, 6, and 7 consisting of relatively fewer data among all surgical steps. Experimental results revealed that surgical phase recognition performance could be improved by applying the adaptive temporal subsampling approach proposed in this study.

When our method and other phase recognition methods in cholecystectomy videos were compared, especially with an architecture such as CNN-LSTM, our model showed better results of phase recognition with the Cholec80 dataset. Temporal information is important for building neural networks for phase recognition of surgical video data. Most existing studies have investigated model structures that connect one convolutional neural network and one LSTM to perform sequence-to-vector [5,19–24,30]. However, our two-stream CNN-LSTMs could leverage sequential information by incorporating different LSTMs. One LSTM was used to capture sequence-to-sequence short-range context information of previous neighboring frames and the other LSTM was used to capture sequence to recognize phase in the last frame of every clip for a long-term view. Experimental results demonstrated that our model could lead to better results of phase recognition using the Cholec80 dataset. These results are due to minimization of lost temporal information by connecting two LSTMs, a sequence-to-vector, and a sequence-to-sequence, in parallel. This has the advantage of additionally learning information on movement changes between image frames as well as overall activity within the short clip.

Although the proposed method showed high performance of surgical phase recognition in cholecystectomy videos, it had four limitations. First, we conducted empirical experiments to determine the optimal conditions of the final model(CNN and the number of image frames in the clip). However, statistical results such as p-values could not be compared because the 15 videos used in the test were insufficient to analyze differences. To enhance the effectiveness of the training, investigating statistical significance under more diverse conditions using a larger dataset could be considered in the future. Secondly, although the overall F1-score and AUC were fairly high, an overall accuracy of 90.47% was considered insufficient for practical applications. To improve the overall performance, a large scale of video focused on true negative and true positive should be used to train the model in future analysis. Thirdly, video clips were used as a set of inputs for our CNN-LSTM models, not the whole video because of memory limitation. To handle the entire video comprising temporal information for all phases, it is necessary to consider other approaches such as temporal convolution or transformer networks in the future. Lastly, our framework requires a fully annotated dataset which can be an annoying task. To reinforce work efficiency, approaches of unsupervised learning or semi-supervised learning, such as the application of generative adversarial networks (GAN), pseudo-labeling, and data augmentation, can be considered in future studies.

6. Conclusion

Surgical workflow recognition in laparoscopic surgical videos is becoming popular with increasing use of minimally invasive surgeries. In particular, surgical phase recognition attempts to make effective use of intra-operative interaction with the surgical team. We proposed a two-stream CNN-LSTM model with short clip-based undersampling for surgical phase recognition in cholecystectomy videos. The proposed method could minimize biased learning by constructing similar numbers of data for each phase after splitting the original surgical video into short clips. We also extracted meaningful temporal information, including both overall action and movement changes between each image frame within a short clip. The experimental results confirmed the excellent performance of the proposed method in surgical phase recognition. In particular, large deviations in the F1-score and AUC of each phase due to the unbalanced structure were significantly reduced. In addition, our two-stream CNN-LSTM structure showed a high F1-score by minimizing the loss of temporal information within a short clip compared to existing studies using a one-stream model. Along with these results, our approach could predict near real-time surgical phases at intervals of one second using a sliding window method. As a result, the proposed adaptive sampling-based undersampling method could effectively minimize overfitting for each phase. The two LSTM models of the short clip-based parallel structure allow learning with more temporal information than conventional models to perform near-real-time surgical phase recognition with high classification performance.

CRediT authorship contribution statement

Sang-Goo Lee: Conceptualization, Methodology, Software. **Ga-Young Kim:** Writing – original draft. **Yoo-Na Hwang:** Writing – review & editing. **Ji-Yean Kwon:** Validation, Visualization. **Sung-Min Kim:** Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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